

Optics Letters

Color domains in fiber lasers

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Received 9 August 2018; revised 14 September 2018; accepted 14 September 2018; posted 17 September 2018 (Doc. ID 341852); published 10 October 2018

We report on the experimental observation of a color domain (CD) phenomenon in a dissipative soliton fiber laser. The CDs are constituted by a wavelength-dependent condensed phase of pulses in motion. Single CD, dual CD, and tricolor domain that occupy all the cavity space are observed. The CDs have flexible tunability of duty cycle, amplitude, wavelength, and wavelength spacing. The CDs derive from the wavelength-dependent slow evolution of gain, and the formation mechanism of multiple CDs can be explained by cross gain saturation. © 2018 Optical Society of America

OCIS codes: (140.3510) Lasers, fiber; (140.4050) Mode-locked lasers; (060.5530) Pulse propagation and temporal solitons.

<https://doi.org/10.1364/OL.43.005054>

Fiber laser cavities are essentially dissipative systems, and passively mode-locked fiber lasers provide a rich platform to study the complex pulse dynamics [1]. Therefore, pulses in such lasers regardless of the dispersion sign can be referred to as dissipative solitons (DS), which are stable and self-localized modes formed in nonlinear dispersive media [2]. Under high pumping power, DS suffer from several instabilities leading to pulse fragmentation or multipulsing. In a normal dispersion regime, it is mainly due to different nonlinear perturbations [3–5], whereas in the anomalous dispersion regime, it is the energy quantization that limits pulse energy and promotes pulse instabilities [6,7]. The energy quantization effect in a normal dispersion regime was also reported [8]. To circumvent these issues, the dissipative soliton resonance (DSR) phenomenon has been proposed as a distinctive pulse mode in the framework of the cubic–quintic Ginzburg Landau equation [9–12]. Under increasing pump power, the DSR phenomenon features the generation of square pulses without wave-breaking [13] due to the peak power clamping (PPC) effect in the cavity. This has been experimentally observed in normal [14,15] and anomalous [16,17] dispersion mode-locked fiber lasers. Meanwhile, the generation of square-shaped pulses is not exclusive to the DSR phenomenon as it has been reported [18–20]. Based on the PPC effect in dispersive resonators, noise-like square pulses (NLPs) have recently been investigated

[21,22]. In contrast with DSR square pulses, which are coherent and highly chirped pulses, an NLP consists of a large soliton packet with random variations of amplitudes and phases under a square envelope [23].

Apart from the DSR phenomenon and NLPs, periodic square profile pulses with a nanosecond time scale have been observed in a polarization resolved erbium-doped fiber laser [24]. Antiphase square-wave pulses and dark pulses in weakly birefringent cavity fiber lasers were also obtained [25–28]. The coherence peak of the autocorrelation trace indicates that there is a fine structure in the polarization resolved square profile pulses [25,27]. A polarization domain wall theory [29–31] has been used to explain the formation of the polarization switch, but the components of the domains and the reason to make them switch are unclear. Further research shows that cross gain saturation (XGS) plays a key role in the formation of polarization domains (PDs) in quasi-isotropic cavity fiber lasers [32], meaning that the gain of the PDs can be naturally self-modulated in a quasi-isotropic cavity.

If a polarization-selective element is inserted into the cavity, multiple-wavelength operation with larger wavelength spacing can be motivated due to the filter effect of the cavity [33]. In this case, square profile domains can also be observed due to the wavelength switch instead of polarization switch [34,35]. Ultimately, questions concerning the components of the domains and the mechanism behind a domain switch can naturally be aroused since there is no straightforward explanation of this behavior.

In this Letter, we demonstrate experimentally that wavelength switch can lead to the generation of wavelength-dependent domains that are formed by a condensed soliton phase. Color domain (CD) is referred to as the domains formed by one central wavelength. The tunability of CD was also investigated. We also establish a relationship between the formation of multi-CD and gain competition dynamics in a single fiber oscillator.

The experimental setup shown in Fig. 1 is based on the same setup used previously [36]. The total cavity length is about 296 m, and the net cavity dispersion is about -6.51 ps^2 . Long cavities favor multiple-wavelength operation and narrow spectral width, which is important for CD operation. The measuring equipment includes a 12 GHz photodiode, a 12 GHz oscilloscope with 40 GS/s, an electronic spectrum analyzer, an

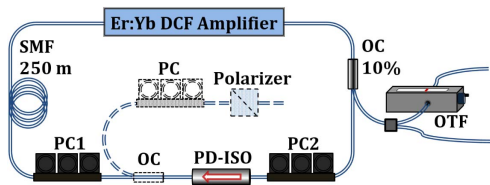


Fig. 1. Experimental setup. DCF, double-clad fiber; OC, output coupler; PC, polarization controller; PD-ISO, polarization-dependent isolator; SMF, single mode fiber; OTF, optical tunable filter.

optical autocorrelator, and an optical spectrum analyzer. The tunable filter (0.5 nm) is used to extract the signals at different wavelengths.

Mode-locking can be easily achieved by increasing the pump power above the threshold and adjusting the polarization controllers (PCs). For a pump power of 0.65 W and by carefully rotating the PC, a large condensed soliton phase referred to as single CD (SCD) is generated. It has one central wavelength located at 1610 nm, and the temporal profile exhibits a square envelope containing fine structure oscillations as confirmed by the autocorrelator. At a fixed orientation of the PCs and by slowly increasing the pump power, the width (duty cycle) and the spectral intensity of the SCDs gradually increase, while the spectral bandwidth and central wavelength remain stable as shown in Fig. 2(a).

The SCD operation is a dissipative regime in which a square envelope of solitons and large background coexist. This derives from a weak mode-locking due to the narrow-band filtering and peak power clamping effect, while the background is formed by the radiation of solitons in motion and other CW components [37,38]. In the background, we observe countless solitons with weak amplitude moving at random positions. In view of the multisoliton gain theory of a fiber laser [39], the mean slow evolution of gain along the cavity can be explained by relating the radiation with high mean gain to the domain and the background to the weaker one. Also, the random weak gain in the domains can be suppressed by the gain saturation effect [40]. When the pump power reaches 2.65 W, the SCD stretches to occupy all the cavity space and duty cycle becomes equal to 1, as shown in the inset of Fig. 2(a). SCD operation at 1566 nm can also be observed when the PCs are altered. Figure 2(b) shows the 1566 nm SCDs as a function of the pump power. The SCDs at 1566 nm have roughly the same characteristics as the SCDs at 1610 nm. Under the SCD operation at 1566 nm, the random gain is suppressed effectively

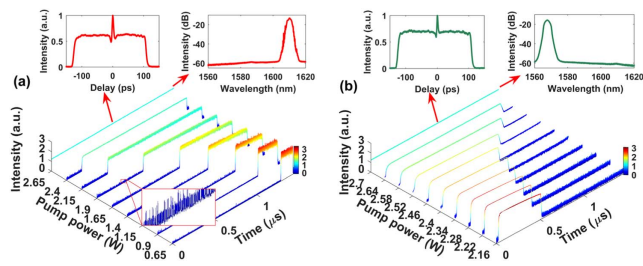


Fig. 2. Temporal profile of the SCD as a function of the pump power. (a) 1610 nm SCD. (b) 1566 nm SCD. Insets of (a) and (b) are the autocorrelation traces and the spectra when SCDs occupy the whole cavity space.

in the domains, which can be reflected from the smoother temporal trace and spectral profile of the SCDs. Insets of Fig. 2 show the autocorrelation traces and the spectra with the SCDs occupying the whole cavity space.

Apart from the effect of the pump power on the SCD operation, we have observed that it can also be influenced by the PC orientation. At a fixed pump power, adjustment of the PC orientation will change the critical saturation power [41]; therefore, the amplitude and duty cycle of the domain will change. Figure 3 shows the SCD variation by adjusting the PC2 orientation with the fixed pump power of 2 W. When we adjust one paddle of PC2 from 0° to 35°, the duty cycle of SCDs increases from 0.6 to 1, while the amplitude and the spectral width decrease. This phenomenon indicates that it is possible to get high duty cycle domains with lower pump power if there is a low enough critical saturation power [23].

It should be noted that the SCD operation and variations are continuous, reversible, and reproducible by controlling the main parameters: the pump power or the PC orientation. The tunable “dark pulse” can be observed in this process. During the SCD operation, we also observed a harmonic distribution of SCD. Then, harmonic “dark pulse” operation can also be observed when the SCDs have a higher duty cycle.

By abruptly altering the PC orientation during an established SCD operation, a CW component at a central wavelength far from the mode-locked wavelength can be observed due to the spectral comb filtering effect of the cavity [33]. When the intensity of the CW component becomes stronger, a large noise background is observed. By finely adjusting the PCs, another wide condensed soliton phase appears from the background. Then, two domains belonging to different wavelengths coexist in the cavity and occupy almost all its space, which is called dual CD (DCD) here. Polarization resolved study on the DCD by using a polarization beam splitter indicates that the DCDs are partially polarized. The polarization resolved spectra are like the results that have been reported [26,28,42], while the polarization resolved temporal traces show that these adjacent domains cannot be completely separated from each other. However, the domains can be completely separated by an optical tunable filter (OTF). To confirm that the formation of the DCDs is not due to the polarization switching, we change the position of OC just after the polarization-dependent isolator (PD-ISO) in the cavity; a PC and a polarizer were also used outside the cavity to confirm the output signal is linearly polarized, as shown in Fig. 1 (dashed part). The same DCD phenomenon is observed in this situation, meaning that the wavelength switch is the key factor to form the DCDs. Figure 4(a) shows the optical spectrum of DCD operation with the pump power of 2.8 W. Figure 4(b) shows the DCD’s temporal traces without a filter and filtered at short wavelength (SW) and long wavelength (LW), respectively. The filtered signals are two antiphase CDs formed by different wavelengths.

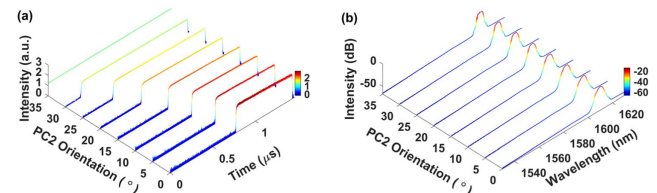


Fig. 3. SCDs influenced by PC2 orientation with fixed pump power. (a) Temporal traces. (b) Optical spectra.

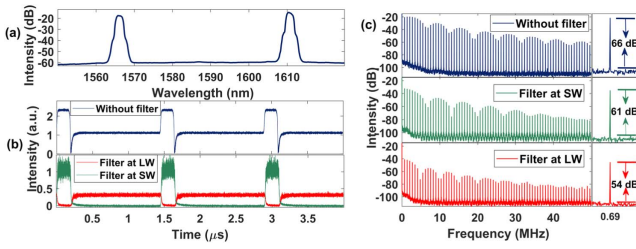


Fig. 4. DCD characteristics. (a) Optical spectrum. (b) Temporal traces of initial and filtered signals. (c) RF spectra.

Highly structured variation of the filtered signal is due to the narrow bandwidth (~ 0.5 nm) of the OTF. Figure 4(c) shows the RF spectra of initial and filtered signals. All the signals have the same repetition rate of 0.69 MHz, which agreed with the cavity length. The periodic modulation of the frequency and the RF signal-to-noise ratio (SNR) of 66, 61, and 54 dB indicates that the distributions are stable.

DCD operation is a dynamic equilibrium, in which two signals compete and exchange energy all the time. This characteristic can be reflected by their inverse duty cycle variation with the PC orientation, as shown in Fig. 5. When one paddle of PC2 orientation is adjusted at a fixed pump power of 2.8 W, the width of one domain increases while the other one decreases. Figures 5(a) (also see Visualization 1) and 5(b) show the initial and filtered signal. Once the signal strength of one wavelength becomes too weak, DCD operation breaks down and SCD operation with a duty cycle of 1 occurs. Regimes with PC2 orientation at 0° and 30° correspond to SCD operations at LW and SW that fill the whole cavity space. The energy transfer between both wavelengths is also indicated by the variation of spectral intensity (linear scale), as shown in Fig. 5(c). The entire experimental process is continuous and reversible by controlling the orientation of PC2. The opposite variation of the duty cycle is shown in Fig. 5(d).

The dynamic equilibrium can also be reflected by the linked temporal intensity variation of both domains with PCs, as shown in Fig. 6(a). When the PC1 and PC2 are adjusted at the same time, the amplitude of one domain increases while the other one decreases. The tunability of the DCD is not limited to the tunable duty cycle and amplitude since different wavelength spacing of DCD operation was also observed in the experiment.

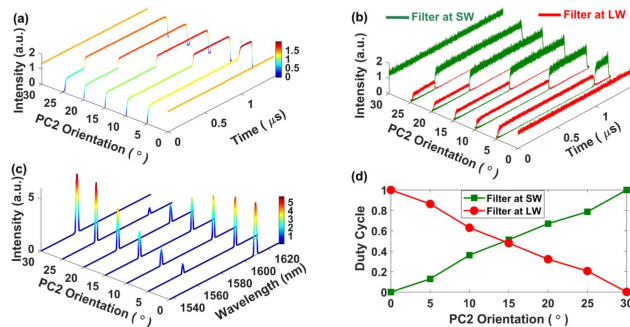


Fig. 5. Tunable characteristic of DCD with PC2 orientations. (a) Temporal traces of the initial output (see Visualization 1). (b) Filtered temporal traces. (c) Optical spectrum. (d) Duty cycle of filtered signals.

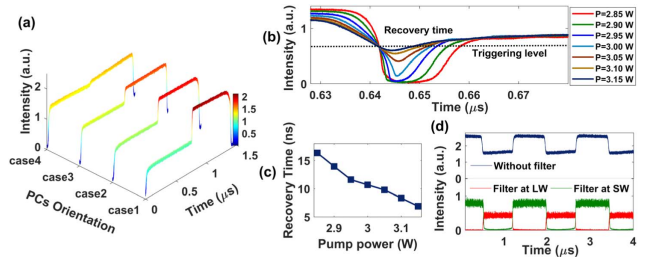


Fig. 6. (a) Tunable amplitude of DCDs with PCs orientations. (b) and (c) Recovery time variation with different pump power. (d) Temporal traces of DCD operation with invisible recovery time.

To understand the reason behind the existence of an intensity dip in a single round trip time under the DCD operation, we investigate the dynamics of this behavior. Since it occurs at the rising or falling edge of the envelope containing both condensed phases, the wavelength switching effect can no longer be considered as a contributing element. At the same time, we have observed dark pulses in the SCD operation, so the intensity dip can be related to the gain depletion and recovery time [39]. To confirm this, we fixed the PC orientation, and we increased the pump power from 2.85 W to the maximum of the amplifier around 3.15 W. We observed that the width of the recovery time (at 80% of the LW amplitude) decreases from 17 to 7 ns, as shown in Figs. 6(b) and 6(c). At the maximum pump power and with adjustment of the PCs, DCDs with invisible recovery time were obtained, as shown in Fig. 6(d). Based on these experimental results, we can relate the existence of dips to the dynamics of gain depletion and recovery time. However, since dips appear only in the DCD regime and not in the SCD regime with the same pump strength as shown in Fig. 5(a), this indicates that the chromatic dispersion induced group velocity difference (walkoff effect) can be related to this behavior [31,37]. In the cavity, the partially polarized characteristic of the DCDs can partly suppress their walkoff effect, but this suppressing effect will disappear when the two signals pass the PD-ISO where there is only one polarized direction. So, the formation mechanism of the intensity dip is the combined effect of gain depletion and chromatic dispersion induced group velocity difference.

The formation mechanism of the DCD is due to the XGS effect of the two domains. When two wide condensed soliton phases at different wavelengths coexist in the cavity, the XGS effect of the two domains make them achieve antiphase gain from the amplifier; then, the two domains appear in different positions due to the self-organizing of the cavity. The dynamic equilibrium of the domains makes them propagate as a unit in the cavity even if there is unsuppressed walk-off effect. When the DCD occupies all the cavity space, a fast cross-gain modulation phenomenon can be considered: two signals transmit in the cavity together; when one signal is considered as a modulating signal, the other signal is modulated to achieve antiphase operation due to the XGS effect. This fast passive cross modulation is based on self-generated mode-locked domain trains of the cavity. It makes the domain switching time exceed the limit of the gain recovery time of the rare-earth doped fibers.

Note that the dual wavelength operation here is different from the doublet spectrum of the DSR phenomenon that we reported recently [36]. The two wavelengths in this experiment

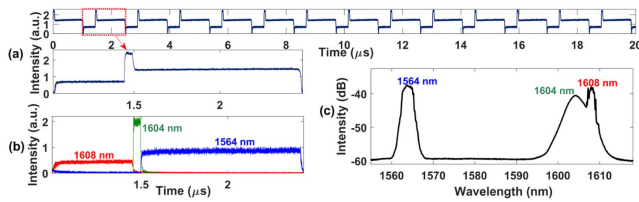


Fig. 7. TCD characteristics. (a) Temporal trace. (b) Filtered temporal traces. (c) Optical spectrum.

have antiphase mean gain to form two incoherent quasi-square pulses, while the two wavelengths of DSR pulses have in-phase gain to form one coherent square pulse in the cavity.

Tricolor domain (TCD) operation was also observed in the experiment, as shown in Fig. 7. The polarization resolved study shows that the three domains are also partially polarized. Figure 7(a) exhibits the temporal trace of TCD operation. The filtered signals at 1608, 1604, and 1564 nm are shown in Fig. 7(b). The corresponding spectrum is presented in Fig. 7(c). The formation mechanism of TCD can be explained in the same way as the DCD, but owing to the complexity of the operational mode, this will be kept for future research. TCD operation has the same tunable behavior of the duty cycle, amplitude, and wavelength spacing as DCD. In this regime, we also observed three-wavelength “dark pulse” and “bright–dark pulse pair,” which were extensively investigated but explained using different approaches.

In conclusion, we have demonstrated a CD phenomenon in a mode-locked fiber laser, and we have investigated its dynamics. To the best of our knowledge, this is the first observation where one or multiple condensed soliton phases can stretch to the full cavity extent to achieve the CD operation. The various “dark pulse” phenomenon can be well understood in this Letter. The essential features of CD can be summarized as (1) a dynamic equilibrium in dissipative systems, (2) tunable duty cycle, amplitude, wavelength, and wavelength spacing, where dark pulses can be observed, (3) possibility to occupy the entire cavity if the pump power is high enough or with a suitable polarization state of the cavity, and (4) the XGS theory as a tool to explain the formation mechanism of multi-CD.

Funding. National Natural Science Foundation of China (NSFC) (11504081); Natural Science Foundation of Hebei Province (A2016208025); Hebei Introduction of Talent Studying Abroad Foundation (CL201712); China Scholarship Council (CSC) (201708130050).

REFERENCES

- P. Grelu and N. Akhmediev, *Nat. Photonics* **6**, 84 (2012).
- N. Akhmediev and A. Ankiewicz, *Dissipative Solitons*, N. Akhmediev and A. Ankiewicz, eds. (Springer, 2005).
- F. W. Wise, A. Chong, and W. H. Renninger, *Laser Photon. Rev.* **2**, 58 (2008).
- P. Grelu and J. M. Soto-Crespo, *Dissipative Solitons: From Optics to Biology and Medicine*, Lecture Notes in Physics (Springer, 2008), Vol. **751**, p. 137.
- M. Liu, A. P. Luo, Y. R. Yan, S. Hu, Y. C. Liu, H. Cui, Z. C. Luo, and W. C. Xu, *Opt. Lett.* **41**, 1181 (2016).
- A. Komarov, A. Haboucha, and F. Sanchez, *Opt. Lett.* **33**, 2254 (2008).
- A. Haboucha, H. Leblond, M. Salhi, A. Komarov, and F. Sanchez, *Opt. Lett.* **33**, 524 (2008).
- X. Liu, H. Wang, Y. Wang, W. Zhao, W. Zhang, X. Tan, Z. Yang, D. Shen, C. Li, Z. Yan, L. Zhang, and G. Chen, *Laser Phys. Lett.* **10**, 095103 (2013).
- W. Chang, A. Ankiewicz, J. M. Soto-Crespo, and N. Akhmediev, *Phys. Rev. A* **78**, 023830 (2008).
- W. Chang, J. M. Soto-Crespo, A. Ankiewicz, and N. Akhmediev, *Phys. Rev. A* **79**, 033840 (2009).
- P. Grelu, W. Chang, A. Ankiewicz, J. M. Soto-Crespo, and N. Akhmediev, *J. Opt. Soc. Am. B* **27**, 2336 (2010).
- E. Ding, P. Grelu, and J. N. Kutz, *Opt. Lett.* **36**, 1146 (2011).
- X. Liu, *Phys. Rev. A* **81**, 053819 (2010).
- X. Wu, D. Y. Tang, H. Zhang, and L. M. Zhao, *Opt. Express* **17**, 5580 (2009).
- Z. Cheng, H. Li, H. Shi, J. Ren, Q. H. Yang, and P. Wang, *Opt. Express* **23**, 7000 (2015).
- K. Krzempek, J. Sotor, and K. Abramski, *Opt. Lett.* **41**, 4995 (2016).
- G. Semaan, F. B. Braham, J. Fourmont, M. Salhi, F. Bahloul, and F. Sanchez, *Opt. Lett.* **41**, 4767 (2016).
- X. Li, X. Liu, X. Hu, L. Wang, H. Lu, Y. Wang, and W. Zhao, *Opt. Lett.* **35**, 3249 (2010).
- X. Zhang, C. Gu, G. Chen, B. Sun, L. Xu, A. Wang, and H. Ming, *Opt. Lett.* **37**, 1334 (2012).
- L. Mei, G. Chen, L. Xu, X. Zhang, C. Gu, B. Sun, and A. Wang, *Opt. Lett.* **39**, 3235 (2014).
- J. Liu, Y. Chen, P. Tang, C. Xu, C. Zhao, H. Zhang, and S. Wen, *Opt. Express* **23**, 6418 (2015).
- X. Li, S. Zhang, M. Han, and J. Liu, *Opt. Lett.* **42**, 4203 (2017).
- Z. S. Deng, G. K. Zhao, J. Q. Yuan, J. P. Lin, H. J. Chen, H. Z. Liu, A. P. Luo, H. Cui, Z. C. Luo, and W. C. Xu, *Opt. Lett.* **42**, 4517 (2017).
- Q. L. Williams and R. Roy, *Opt. Lett.* **21**, 1478 (1996).
- H. Zhang, D. Tang, L. Zhao, and X. Wu, *Phys. Rev. B* **80**, 052302 (2009).
- H. Zhang, D. Y. Tang, L. M. Zhao, and R. J. Knize, *Opt. Express* **18**, 4428 (2010).
- C. Lecaplain, P. Grelu, and S. Wabnitz, *J. Opt. Soc. Am. B* **30**, 211 (2013).
- X. Li, S. Zhang, H. Han, M. Han, H. Zhang, L. Zhao, F. Wen, and Z. Yang, *Opt. Express* **23**, 10747 (2015).
- M. Haelterman and A. P. Sheppard, *Opt. Lett.* **19**, 96 (1994).
- S. Pitois, G. Millot, and S. Wabnitz, *Phys. Rev. Lett.* **81**, 1409 (1998).
- D. Tang, Y. Song, J. Guo, Y. Xiang, and D. Shen, *IEEE J. Sel. Top. Quantum Electron.* **20**, 42 (2014).
- C. Lecaplain, P. Grelu, and S. Wabnitz, *Phys. Rev. A* **89**, 063812 (2014).
- Z. Luo, A. Luo, W. Xu, H. Yin, J. Liu, Q. Ye, and Z. Fang, *IEEE Photon. J.* **2**, 571 (2010).
- H. Zhang, D. Tang, L. Zhao, and X. Wu, *Opt. Express* **19**, 3525 (2011).
- Z. Lin, A. Luo, S. Wang, H. Wang, W. Cao, Z. Luo, and W. Xu, *Opt. Laser Technol.* **44**, 2260 (2012).
- G. Semaan, A. Komarov, A. Niang, M. Salhi, and F. Sanchez, *Phys. Rev. A* **97**, 023812 (2018).
- S. Chouli and P. Grelu, *Phys. Rev. A* **81**, 063829 (2010).
- C. Bao, X. Xiao, and C. Yang, *Opt. Lett.* **38**, 1875 (2013).
- A. Haboucha, H. Leblond, M. Salhi, A. Komarov, and F. Sanchez, *Phys. Rev. A* **78**, 043806 (2008).
- A. Komarov, F. Amrani, A. Dmitriev, K. Komarov, and F. Sanchez, *Phys. Rev. A* **87**, 023838 (2013).
- Y. Jeong, L. A. Vazquez-Zuniga, S. Lee, and Y. Kwon, *Opt. Fiber Technol.* **20**, 575 (2014).
- P. Wang, K. Zhao, X. Xiao, and C. Yang, *Opt. Express* **25**, 30708 (2017).